Power Input, Buck, Charger & Power Path

File: power.kicad_sch

Cat-1 / GSM Modem, SIM & LTE RF

File: modem.kicad_sch

GNSS (L76K) & Level Shift

File: gnss.kicad_sch

Accelerometer, Vehicle I/O & LEDs

File: io.kicad_sch

CRL-TRK-2W-L1 Two-wheeler tracker LITE – 60 x 45 mm, 2-layer 1.6 mm FR-4, single-sided SMT

VEHICLE 9-30 V -> F1 / D1 TVS -> D2 reverse diode -> XL1509 buck 4.44 V -+-> D4 -> VSYS 3.3-4.3 V -> EC200U VBAT
+-> MCP73831 -> 300 mAh LiPo -> Q1 P-FET -> VSYS

EC200U-CN (Cat-1 bis + GSM, runs the firmware) -> U.FL -> FPC LTE/GSM antenna
<-MAIN UART (BSS138 1.8/3.3 V)-> L76K GPS/BDS/GLONASS -> U.FL -> active L1 patch
<-I2C-> LIS3DH <-GPIO-> IGN sense, relay driver, GNSS_EN <-ADC-> VIN/30, VBAT/4

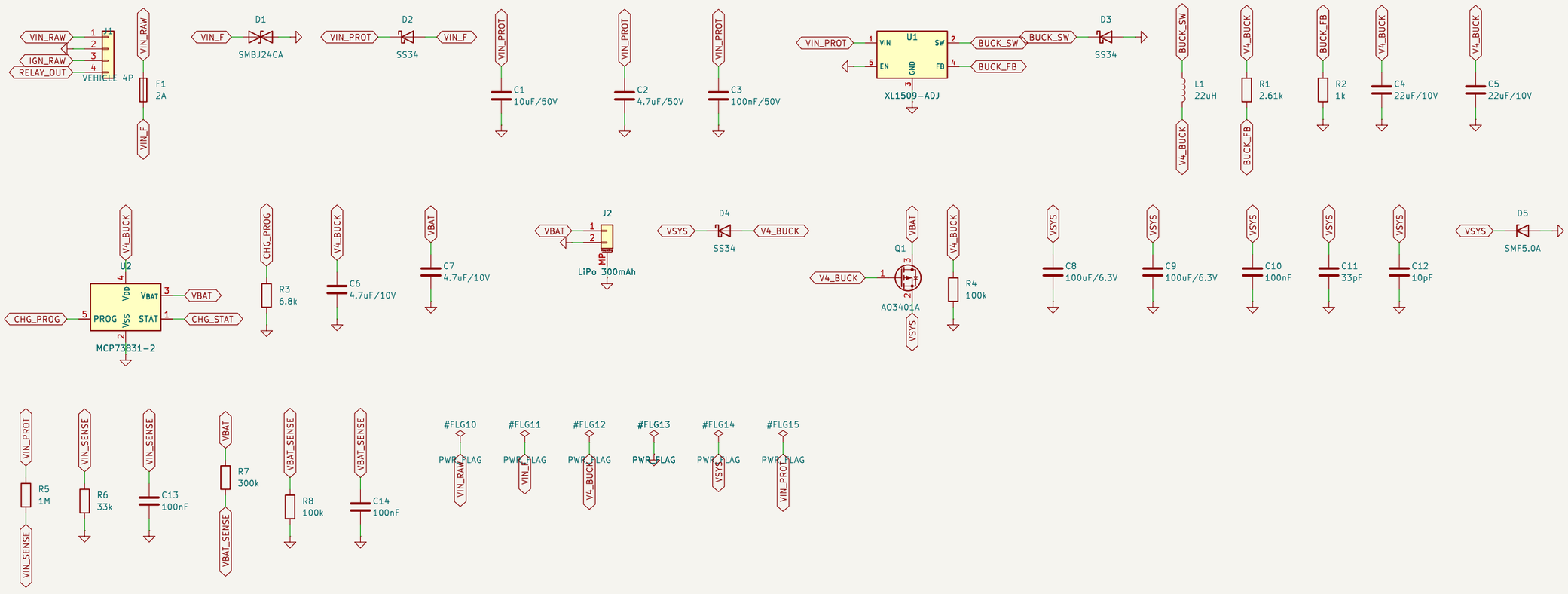
Parked (ignition off): modem sleep + GNSS off + LIS3DH wake-on-motion, ~2-3 mA from the vehicle battery.

Top level: block diagram
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Sheet: /
File: tracker_lite.kicad_sch

Title: CRL-TRK-2W-L1 – Top level

Size: A3	Date: 2026-10-20	**Rev: A0**
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DESIGN NOTES
Input 9–30 V (12 V two-wheelers): F1 fuse, D1 SMBJ24CA load-dump clamp, D2 SS34 series reverse diode.
XL1509 40 V / 2 A non-sync buck -> V4_BUCK = 4.44 V (R1/R2). It carries the 2 A GSM bursts.
MCP73831 charges the 300 mAh LiPo from V4_BUCK at 147 mA (tops out ~4.1 V: longer cell life).
Power path: V4_BUCK -> D4 -> VSYS (4.1 V); Q1 P-FET connects the battery to VSYS when the buck stops.
VSYS (3.3–4.3 V) feeds the EC200U VBAT_BB/VBAT_RF directly: 220 uF + 100 uF + 100n/33p/10p + 5 V TVS.
Layout: buck loop C2–U1–D3 short, SW copper small; keep the buck away from the GNSS corner.

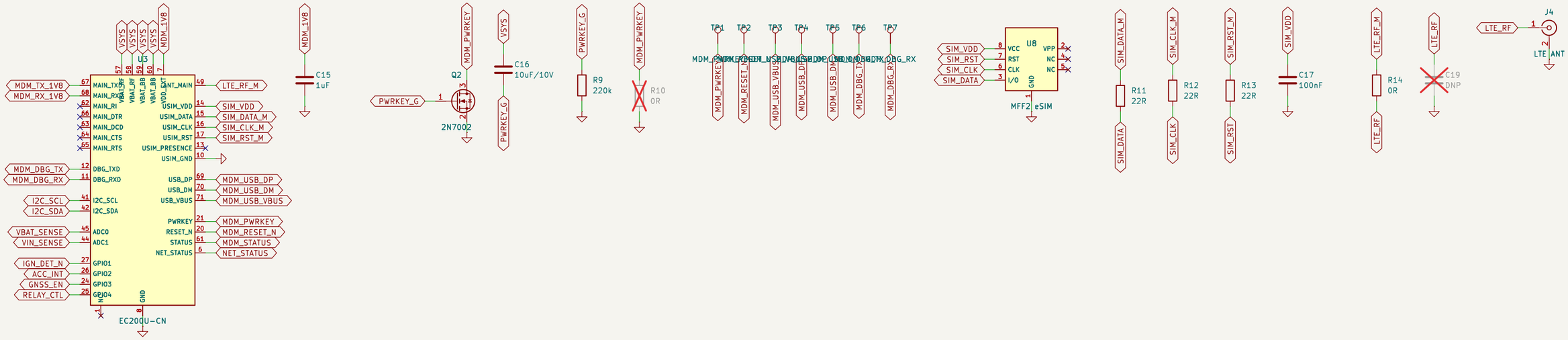
PART NOTES (VERIFY = check against vendor datasheet before release)
J1: 1 +12V(BAT) 2 GND 3 IGN 4 RELAY (low side, coil to +12V)
D1: clamp 38.9 V < XL1509 40 V abs max: reverse battery blows F1 via D1
C1: input bulk; the harness inductance is damped by D1 + F1
U1: EN (pin 5) low = on, tied to GND; pins 6–8 GND heat-spreader – VERIFY pinout vs XLSEMI datasheet
L1: VERIFY Isat >= 2.2 A (GSM burst); alt. CYNTEC / Sunlord 6045 22 uH
R1: VOUT = 1.23 V x (1 + R1/R2) = 4.44 V
U2: Input is the 4.44 V buck rail: the cell tops out ~4.1 V (longer life in a hot under-seat box)
R3: ICHG = 1000 V / R = 147 mA (0.5 C for 300 mAh)
Q1: Gate = buck rail: off while the buck runs (Vgs = +0.35 V), on (Vgs = -VBAT) when it stops
R4: discharges the buck rail so Q1 turns on fast
C8: C8 + C9 = 200 uF ceramic GSM burst reservoir (~120 uF at 4 V bias)
R6: VIN 0–37 V -> 0–1.18 V (ADC1)
R8: VBAT/4 -> ADC0, 10 uA

Sheet 2: Power Input, Buck, Charger & Power Path
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Sheet: /Power Input, Buck, Charger & Power Path/
File: power.kicad_sch

Title: CRL–TRK–2W–L1 – Power Input, Buck, Charger & Power Path

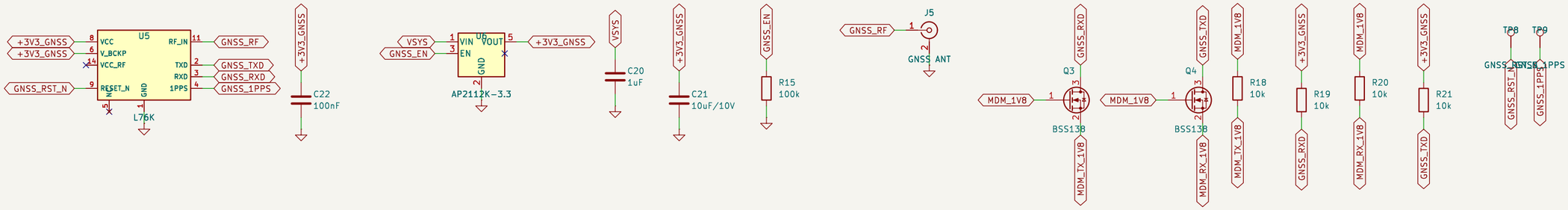
Size: A3	Date: 2026-10-20	**Rev: A0**
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DESIGN NOTES
EC200U-CN runs the tracker firmware (QuecOpen / QuecPython): no separate MCU.
Auto power-on: VSYS step charges C16 -> Q2 pulls PWRKEY low ~1.5 s. R10 (DNP) = permanent auto-on option.
Firmware download and logs through the modem USB (TP3-TP5) and debug UART (TP6/TP7) on a pogo fixture.
SIM: soldered MFF2 M2M eSIM (U8), 22 R series. LTE RF: 50 R CPWG -> pi pad (C18/R14/C19) -> U.FL.
LAND PATTERN U3 = parametric reconstruction - replace with the Quectel recommended footprint (VERIFY_LIST).

PART NOTES (VERIFY = check against vendor datasheet before release)
U3: Runs the tracker firmware (QuecOpen / QuecPython). Land pattern is a reconstruction - VERIFY
C16: power-on pulse: tau = C16 x R9 = 2.2 s, PWRKEY low >= 500 ms
R10: fit for permanent auto power-on (no PWRKEY shutdown)
U8: Pin-out per ETSI TS 102 671 MFF2 (1 GND 2 VPP 3 I/O 6 CLK 7 RST 8 VCC) - VERIFY with the SIM vendor
C19: L-match with R14: shunt C or L on the bench

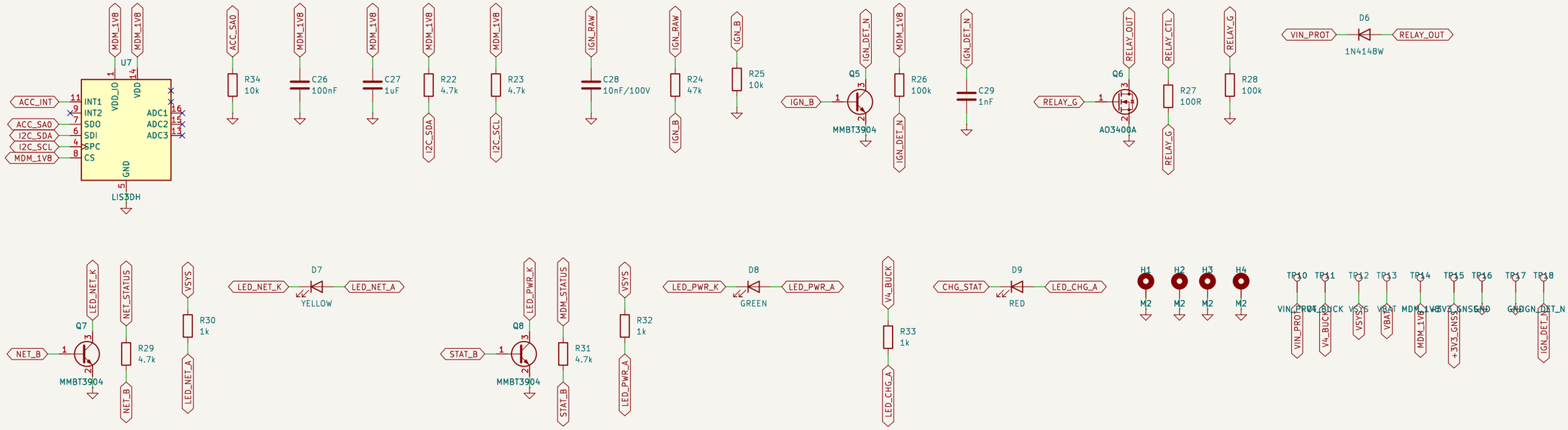
Sheet 3: Cat-1 / GSM Modem, SIM & LTE RF
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Sheet: /Cat-1 / GSM Modem, SIM & LTE RF/
File: modem.kicad_sch
Title: CRL-TRK-2W-L1 - Cat-1 / GSM Modem, SIM & LTE RF
Size: A3 | Date: 2026-10-20 | **Rev: A0**
KiCad E.D.A. 9.0.9 | Id: 3/5



DESIGN NOTES
L76K: GPS + BeiDou + GLONASS + QZSS, L1 only (no NavIC; Lite is not for AIS-140).
U6 LDO switched by modem GPIO3 (GNSS_EN): GNSS fully off when parked. V_BCKP = VCC.
UART 1.8 V <-> 3.3 V through Q3/Q4 BSS138 shifters (modem MAIN UART = NMEA port).
Active patch bias from VCC_RF via R16/L2; DNP R16 + L2 for a passive patch. RF_IN direct to the U.FL.

PART NOTES (VERIFY = check against vendor datasheet before release)
U5: V_BCKP tied to VCC: hot start only while GNSS_EN is high (AGNSS on cold start) — VERIFY pins vs L76K HD

Sheet 4: GNSS (L76K) & Level Shift
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Sheet: /GNSS (L76K) & Level Shift/
File: gnss.kicad_sch
Title: CRL-TRK-2W-L1 — GNSS (L76K) & Level Shift
Size: A3 | Date: 2026-10-20 | **Rev: A0**
KiCad E.D.A. 9.0.9 | Id: 4/5



DESIGN NOTES
LIS3DH on VDD_EXT 1.8 V, I2C 0x18, INT1 -> GPIO2: wake-on-motion, tow / theft and fall alerts.
Ignition: 47k/10k into Q5, IGN_DET_N active low on GPIO1 (wake-capable). Survives 0-60 V and reverse.
Relay output: Q6 low-side driver for an immobiliser relay coil, flyback D6 to VIN_PROT.
LEDs: green = modem on (STATUS), yellow = network (NET_STATUS), red = charging (MCP73831 STAT).

PART NOTES (VERIFY = check against vendor datasheet before release)
U7: Runs from VDD_EXT 1.8 V, no level shifting
R34: SDO/SA0 low -> I2C address 0x18
R24: IGN > ~3.5 V reads as ON
Q6: Gate from 1.8 V GPIO: Rds(on) ~0.1 R at 1.8 V - VERIFY for the coil current

Sheet 5: Accelerometer, Vehicle I/O & LEDs
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Sheet: /Accelerometer, Vehicle I/O & LEDs/ File: io.kicad_sch		
Title: CRL-TRK-2W-L1 - Accelerometer, Vehicle I/O & LEDs		
Size: A3	Date: 2026-10-20	**Rev: A0**
KiCad E.D.A. 9.0.9		Id: 5/5